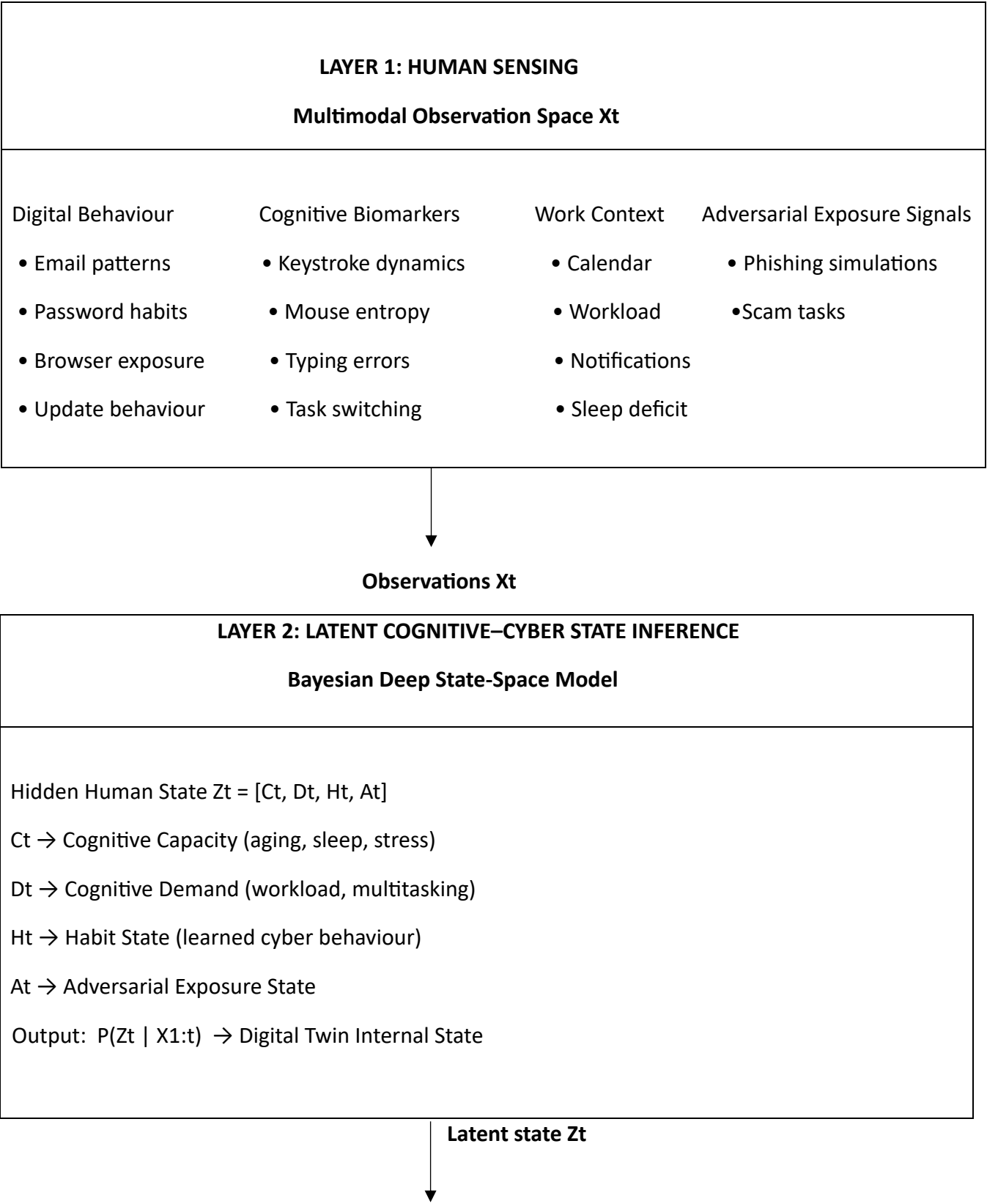
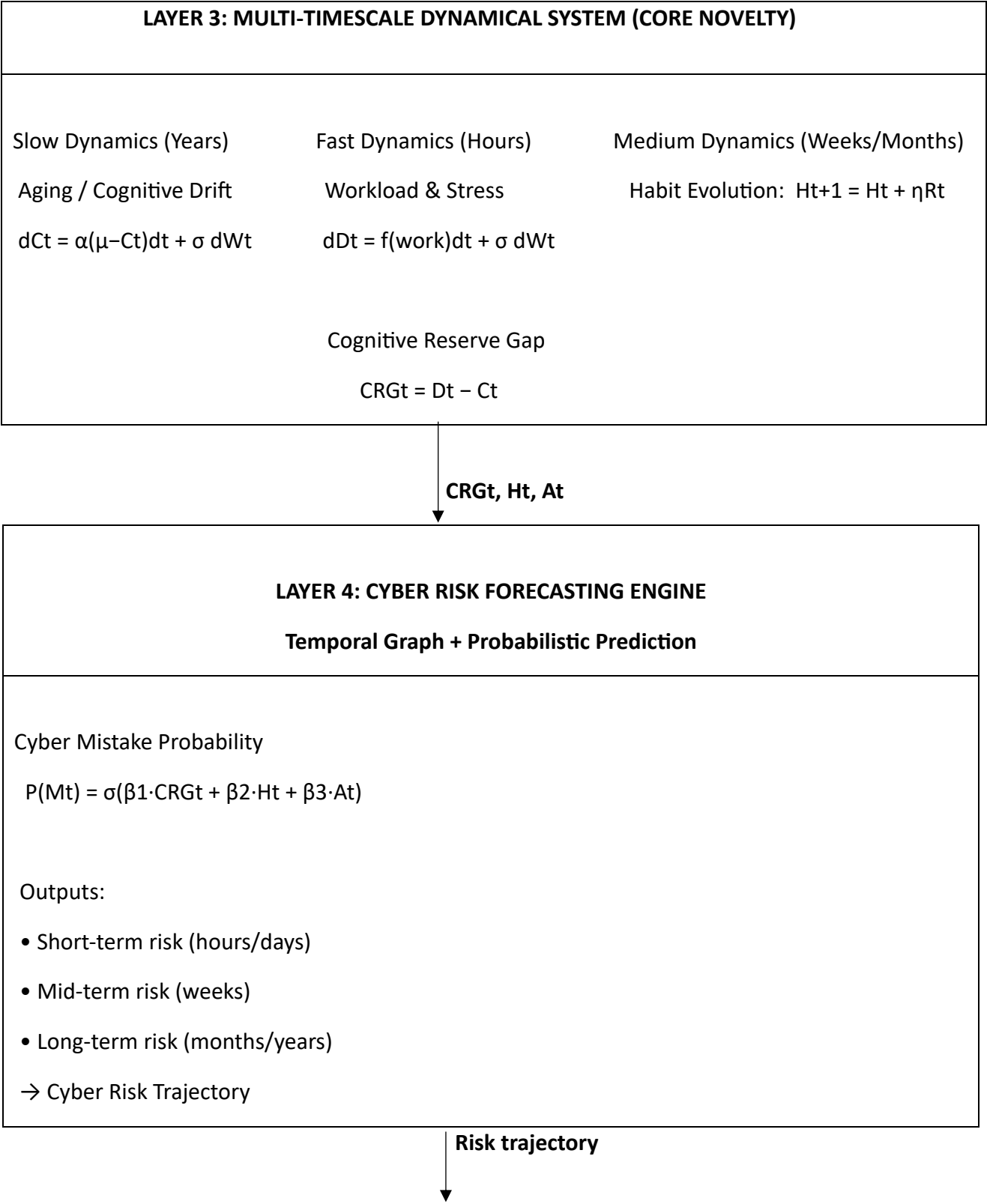


Lifelong Cognitive–Cyber Digital Twin (LCDT)





LAYER 5: DIGITAL TWIN SIMULATION ENGINE
Generative & Counterfactual Modeling

Simulates Future Scenarios:

- Increased workload
- Aging progression
- Security training
- UI / policy changes

Counterfactual Estimation

$$E[Mt:T \mid do(\pi)]$$

Output: Future Cyber-Risk Forecasts